

jigs, and end-use components too. The energy sector is using additive manufacturing to print parts for customised solutions, while carmakers are using it to customise luxury cars. The aerospace world is using it extensively to print lightweight but sturdy parts and components from special materials, while the food industry is using it to make customised confectioneries and personalised nutrition solutions.

From construction to electronics, everyone seems to be exploring how they can utilise the potential of 3D printing. Let us share some of those insights.

Print a pill? Old news! Ready to print organs now

According to a report by Grand View Research, an India and the US based market research and consulting company, the global healthcare additive manufacturing market size was valued at US\$1.6 billion in 2021. It is expected to expand at a compound annual growth rate (CAGR) of 22.6% from 2022 to 2030. The need for customised medical equipment like implants, the emergence of new 3D printing technologies that can economically print complex designs, and development of new biocompatible material capable of printing tissues, organs, dental fittings, orthopaedic and cranial implants are contributing to this growth.

Medical equipment like implants is very expensive to produce with traditional equipment, moulds, dies, and all the works. After all, most of these are comparable to works of art, tailor-made to fit a person. On top of that, these have to be produced quickly; sometimes it is a matter of life and death. The ability to swiftly develop prototypes using computer-aided



Crescent, the Great Indian Hornbill at ZooTampa at Lowry Park, got a new 3D-printed casque to replace her cancer-ridden one (Courtesy: ZooTampa, Formlabs)

design (CAD) makes 3D printing fast and economical.

Helping hand during the pandemic. 3D printing rose to the occasion to support healthcare systems during the Covid-19 pandemic. Lock-downs crippled supply chains, but there was a growing need for supplies like personal protective equipment (PPE). 3D printing enthusiasts from around the world, university labs, and companies that had 3D printers started meeting this demand by printing goods like face shields and valves for respiratory machines. For example, 3T Additive Manufacturing Ltd and EOS produced more than 100,000 face shields for healthcare workers in the UK. Allevi Inc, a 3D bio-fabrication company, produced a model of the lungs of a person infected with SARS-CoV-2, which was studied by researchers at the Wistar Institute to understand the condition and find ways to treat and prevent it.

Prosthetics and exoskeletons. Earlier this year, 3D printing saved

the life of a baby girl born with a critically-underdeveloped cranium—in simple terms, without a skull. A doctor approached Sygnis, a Polish deep tech company, just hours before the baby was due to be born. Armed with imaging data, Sygnis started 3D printing the skull using two different technologies, in case one failed. Within a day, they had the model ready—and the doctors completed fitting it within four days. The baby will require further treatment as she grows, and hopefully tech will find a way to help her survive.

Recently, Sygnis also produced another pre-operative model—the spine of a young child with meningo-spinal hernia and scoliosis.

3D printing is also proving to be very effective in the manufacture of exoskeletons for rehabilitation. 3D-printed exoskeletons are lighter, more durable, lower in cost, and a perfect fit for the person.

There are also some open source projects, which aim to make low-cost 3D-printed prosthetics available to underserved populations across the world. Cyborg Beast, for example, is a low-cost 3D-printed prosthetic hand for children with upper-limb reductions. The designs are available under a Creative Commons license, so anyone can download, customise, and 3D-print them.

According to the Biomechanics Department of the University of Nebraska Omaha, the majority of the required materials are available at local hardware stores or online, costing approximately US\$50. It takes around 2.5 hours to assemble the prosthetic hand, which weighs less than 200 grams. A similar device (made using traditional methods) would cost approximately US\$4000 and weigh about 400 grams.

Blood vessels, tissues, and organs too. The emerging space of 3D bio-printing involves printing with cells and biomaterials, creating organ-like structures that let living cells multiply. These tissues and organ models help

3D Printing: A Quick Background

3D printing, or additive manufacturing, is a technology that has been developing slowly and steadily since the 1980s. In simple terms, a 3D printer can take CAD file as input and print a three-dimensional object by depositing layer after layer of a suitable material. Hence, it is known as 'additive' manufacturing. In some rare cases that require extreme precision, a subtractive process might also take place, where excess material is shaved off the object.

Over the years, several technologies have emerged, capable of printing with different materials, and with varying degrees of precision. Each generation, or each new process, aims to bring down the manufacturing time and costs, increase the precision, and widen the spectrum of materials that can be used. Now, materials ranging from gypsum and ceramic to wood and titanium are being used for 3D printing. As a result, the potential applications are also increasing.